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# DEVELOPMENT OF A ROAD MORTALITY HOTSPOT MODEL FOR AMPHIBIANS AND REPTILES IN RHODE ISLAND

Noah Hallisey
*University of Rhode Island*, halliseyn@uri.edu

DEVELOPMENT OF A ROAD MORTALITY HOTSPOT MODEL FOR

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BY

NOAH HALLISEY

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NOAH HALLISEY

APPROVED:

Thesis Committee:

Major Professor Nancy Karraker

Scott Buchanan

Andy Davies

Brian Gerber

Brenton DeBoef
Dean of the Graduate School

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## ABSTRACT

Road mortality poses a serious threat to amphibians and reptiles and can lead to population declines and localized extirpation. Given road mortality tends to aggregate spatially at locations commonly referred to as road mortality hotspots, identifying the most appropriate locations for mitigation measures, such as road tunnels and culverts, is an important first step in reducing road mortality. However, the influence of imperfect detection (e.g., false absences) during road mortality surveys can lead to poor spatial patterns of road mortality hotspots and suboptimal implementation of mitigation measures intended to reduce road mortality. In this study, occupancy modeling was used to identify where large amphibian and reptile roadkill events (≥ 5 carcasses) occurred along transects in Rhode Island, USA. I defined occupancy as the likelihood of a large roadkill event occurring at a transect. I defined detection probability as the probability of a large roadkill event being detected during a survey assuming it occurs at a transect. From 2019 to 2021, road mortality surveys were conducted along 48, 200-m long transects on 50 occasions for a total 240 surveys. A total of 606 carcasses were observed during surveys representing 19 of Rhode Island's native amphibian and reptile species. I developed and tested models to examine factors associated with large roadkill events. The most supported model estimated a higher occurrence of large roadkill events on roads with a high proportion of wetlands (forest and non-forested wetlands) within 100-m and water bodies (streams and lakes) within 500-m and on roads with low traffic volume (15–400 vehicles/day). For detection probability, large roadkill events were more likely to be detected as daily precipitation increased and less likely to be detected on roads with low traffic volume (15–400 vehicles/day). Large roadkill events were observed at 40% of transects during surveys. However, model results indicated that large roadkill events

likely occurred at 64% of transects and there was a 17.5% chance of detecting a large roadkill event at transects where at least one occurred. The low detection probability suggest that imperfect detection influences the ability to detect large roadkill events, and many are likely to be missed with regularity using our methodology. To better identify road morality hotspots and target locations for mitigation measures, I recommend surveys be conducted on low traffic roads near wetlands. In addition, I recommend surveying during or after a rain event, as the probability of detecting a large roadkill event increases with rain. The approach I have developed can be used to guide survey efforts and address imperfect detection during surveys to better identify locations most appropriate for the implementation of mitigation measures that will be effective in reducing amphibian and reptile road mortality.

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## INTRODUCTION

As road infrastructure that connects cities and rural areas in the United Sates increases, so do the negative impacts of roads and vehicle traffic on ecosystems and wildlife. Among impacts, the most notable is the contribution of roads and traffic to large losses of wildlife through direct mortality (Hill et al., 2019). It has been estimated that between 89 and 340 million birds are struck and killed on roads annually in the United States (Loss et al., 2014). Among vertebrates, road mortality exerts significant impacts on amphibians and reptiles. Herpetofauna play important ecological roles, acting as both predator and prey, supporting energy transfer between aquatic and terrestrial ecosystems, and are also considered indicator species for the health of aquatic ecosystems (Guyer & Bailey, 1993; Welsh Jr & Ollivier, 1998). With amphibian and reptile populations declining worldwide, there is concern surrounding the exacerbation of these declines caused by road mortality (Fahrig et al., 1995; Gibbs & Shriver, 2005). Population declines associated with human-caused stressors, such as road mortality, can disrupt food webs and ecosystem function (Karraker et al., 2020).

Road mortality predominantly impacts herpetofauna, with amphibians accounting for 60–90% of all roadkill observations (e.g., Glista et al., 2008; Matos et al., 2012). Herpetofauna possess ecological and behavioral characteristics that make them highly susceptible to road mortality. Amphibians make frequent road crossings during annual migrations between habitats to breed and forage, increasing the risk of road mortality (Colino-Rabanal & Lizana, 2012; Hels & Buchwald, 2001). For example, the northern leopard frog (*Lithobates pipiens*) regularly crosses roads when moving between upland and wetland habitats to breed and forage (Bouchard et al., 2009). During movements to nest sites, female turtles cross roads and will nest in the loose sandy areas adjacent to roads, increasing the risk of road mortality (Steen et al., 2006). Many snake species overwinter communally and their spring emergence, in which they may leave hibernacula by the hundreds, may result in more frequent road crossings (Jochimsen et al., 2014).

For both amphibians and reptiles, periods of increased risk of road mortality exist during particular times of year and under specific weather conditions tied to a species' natural history and phenology (Beaudry et al., 2010; Cureton & Deaton, 2012). For example, in Indiana (USA), increased levels of road mortality were observed in the summer months from May through July, during periods of time when many species are migrating to breed and forage (Glista et al., 2008). For amphibians, increased road mortality has been observed on evenings with warmer temperatures and precipitation during the breeding season (Gravel et al., 2012; Zhang et al., 2018). For reptiles, high risk for road mortality occurs in the spring and early summer months when many reptile species are frequently crossing roads to reach nesting sites to lay eggs (Beaudry et al., 2010) and emerge in masses from overwintering sites (Jochimsen et al., 2014).

Road surfaces constructed of asphalt and concrete absorb and retain heat, attracting snakes, frogs, and salamanders to use roads to bask and thermoregulate (Andrews & Gibbons, 2005; Gibbs & Shriver, 2005). For diurnal species, heat absorbed by roads on sunny days make road surfaces ideal for basking. Heat absorbed by roads during the day is retained and can radiate into the evening hours, serving as an important source of heat for nocturnal species on cool nights. For both diurnal and nocturnal species, this use of roads can increase the risk of road mortality. Many amphibians move slowly and some species may remain immobile in response to traffic, which can increase the risk of being killed (Brehme et al., 2018; Mazerolle et al., 2005). In addition, many amphibians are small, making them hard for drivers to see and avoid (Brehme et al., 2018). Some snake species and other reptiles are targeted by drivers (Crawford & Andrews, 2016; Ashley et al., 2007).

Road mortality can ultimately lead to population declines in amphibians and reptiles (Colino-Rabanal & Lizana, 2012). This can alter the demographic and genetic composition of populations and fragment breeding populations (Carr & Fahrig, 2001; Gibbs & Shriver, 2005), which can further exacerbate population declines and lead to extirpation (Steen & Gibbs, 2004). For example, an annual road mortality greater than 10% for adult spotted salamanders (*Ambystoma opacum)* was identified as the threshold that could lead to local population declines and potentially extirpation of populations (Gibbs & Shriver, 2005). For terrestrial and large-bodied turtles in the northeastern United States, an annual road mortality greater than 5% could threaten local populations (Gibbs & Shriver, 2002). For turtles, life history traits, such as delayed maturity and low fecundity, make them highly susceptible to the impacts of road mortality (Howell & Seigel, 2019).

Road mortality does not occur randomly but is often spatially clustered at locations with specific landscape and road features, commonly referred to as road mortality hotspots (Clevenger et al., 2003). For herpetofauna, the presence of habitats, the proximity of habitats to roads, and traffic volume are strong indicators of locations with high levels of road mortality (Glista et al., 2008; Matos et al., 2012; Orłowski et al., 2008). For example, in one study (Langen et al., 2009), sections of roads within 100 meters of a wetland was determined to be the most important indicator of amphibian and reptile road mortality hotspots. For amphibians, road mortality hotspots occur at sections of roads with wetlands directly adjacent to or bisected by roads (Patrick et al., 2012). For turtles, sections of road within close proximity to bodies of water were the best predictor of road mortality hotspots (Langen et al., 2012). Road features, including road width and traffic volume, are also important predictors of road mortality hotspots. Wide roads can have multiple lanes to support high levels of traffic and can take longer for herpetofauna to cross, increasing the chance of being killed (Gu et al., 2011). Traffic volume, a measure of the number of vehicles on a road over a given period, usually daily, is another important road feature associated with road mortality hotspots. Although the risk of being killed increases with traffic volume (Hels & Buchwald, 2001), some studies have observed the highest levels of road mortality on roads with lower traffic volume. For example, roads with low (350–470 vehicles/day) and moderate (1900–2900 vehicles/day) traffic volume had the highest levels of amphibian mortality in Poland (Orłowski et al., 2008). Roads with lower traffic volume have less vehicle disturbance for animals, making them more likely locations for thermoregulating amphibians and reptiles, thus increasing the risk of being killed (Colino-Rabanal & Lizana, 2012). In addition, roads with the highest levels of traffic volume that have been in use for decades, such as major interstates and highways, may have initially had higher levels of road mortality when first constructed. However, over time the high levels of road mortality caused populations near the road to experience declines leading to decreases in road mortality over time (Fahrig et al., 1995; Sutherland et al., 2010). Roads with higher traffic volume tend to also be in highly developed areas with less habitat for amphibians and reptiles. High traffic volume may also deter some species from attempting to make a road crossing (Jacobson et al., 2016).

Road surveys are used to identify road mortality hotspots by recording live and dead individuals on roads during high risk periods of road mortality (Langen et al., 2007). Common survey methods include driving or walking along roadways. Driving surveys can be used along large road networks, but this method results in lower detection rates and missed carcasses on the road, underestimating road mortality levels (Langen et al., 2007). For example, it was estimated that the number of roadkill observed during driving surveys is 12–16 times lower than the actual mortality rate (Slater, 2002). Walking surveys have higher detection rates and generate better estimates of road mortality but are time consuming and cover less roadway than driving surveys (Colino-Rabanal & Lizana, 2012; Langen et al., 2007). A recent study comparing methods observed that 75% of amphibian carcasses recorded during walking surveys were missed during driving surveys (Ogletree & Mead, 2020). The frequency of surveys is also important and can impact hotspot identification, with weekly or longer intervals between surveys reducing the accuracy of identifying hotspots for amphibians and reptiles (Santos et al., 2015). Regardless, techniques commonly used to identify hotspots rely on counts of roadkill recorded during surveys, which are often assumed to be underestimated due to imperfect detection during surveys (Glista et al., 2008; Teixeira et al., 2013).

Since large roadkill events tend to occur on only a few nights during the year and under certain weather conditions, the timing of surveys is important for capturing peaks in road mortality, especially since carcasses may remain on the road for brief periods following a large event (Degregrio et al., 2011). In other words, there may be a high risk of road mortality along a section of road, however, due to the timing of the survey, a low level of roadkill is observed. This leads to missing road mortality hotspots. Several studies focusing on herpetofauna road mortality have recognized that the number of roadkill they observed during surveys was likely underestimated due to imperfect detection during surveys which likely influenced spatial patterns of road mortality (Glista et al., 2008; Zhang et al., 2018). Imperfect detection can lead to the false absence (or presence) of hotspots and spatial bias in road mortality patterns, guiding management efforts and mitigation measures that are ineffective in reducing road mortality (Santos et al., 2018; Santos et al., 2015; Teixeira et al., 2013).

Predictive models can be used to identify road mortality hotspots, as well as potential hotspots not previously surveyed, by investigating relationships between numbers of animals killed on roads and particular landscape and road features (Langen et al., 2009; Malo et al., 2004). Studies have used spatial clustering techniques, such as Getis-Ord-Gi (Garrah et al., 2015; Healey et al., 2020; Shilling & Waetjen, 2015), to identify clustering of roadkill to identify road mortality hotspots. However, techniques commonly used rely on roadkill count data and do not address the influence of imperfect detection during surveys on roadkill counts, which can lead to false spatial patterns of road mortality hotspots. Occupancy models, a predictive modeling approach that considers detection probability to estimate occurrence –the probability of a species being present at a location but not detected during surveys – can be used to overcome imperfect detection during surveys to more precisely predict spatial patterns of road mortality (MacKenzie et al., 2017; Santos et al., 2018; Teixeira et al., 2013). Occupancy models consider site characteristics (e.g., the presence of a species' habitat) as well as the environmental conditions during surveys (e.g., precipitation) to determine the probability of a species being present at a location and observed during a survey (i.e., detection or non-detection of a species). Applied to wildlife road mortality, occupancy is a proxy for determining the risk of road mortality, and detection probability is the likelihood of a carcass being detected during a survey, given that it occurs (Santos et al., 2018). Estimating detection probability requires multiple visits to a site and since multiple surveys may be required to identify a road mortality hotspot, the sampling design used for road surveys can be integrated to correct for imperfect detection during surveys (Santos et al., 2018). As occupancy models take into account detection probability, they can be used to correct for false absences by identifying locations with the greatest risk of road mortality rather than identifying locations with the highest observations of roadkill, guiding mitigation measures to locations where road mortality is most likely to occur (Santos et al., 2018). Occupancy detection models are commonly used to estimate site occupancy and species distribution and are used to monitor wildlife populations (Pavlacky et al., 2012; Pellet & Schmidt, 2005). However, few studies have used occupancy detection models to address imperfect detection during road mortality surveys with the goals of improving identification of locations with the greatest risk of road mortality and enhancing recommendations for locations where mitigation measures may be most appropriate in reducing road mortality (Santos et al., 2018).

New England states (U.S.A.), including the state of Rhode Island, are characterized by ongoing development, including road construction, and high human population densities. Road mortality is likely a major problem for native amphibian and reptile populations in Rhode Island, but this anthropogenic impact has not been previously studied. The objectives of my study were to: (1) use an occupancy modeling framework to identify landscape and road characteristics most strongly associated with road mortality hotspots; and (2) make recommendations for enhancing current modeling and survey techniques to identify and prioritize locations for mitigation measures that reduce amphibian and reptile road mortality.

## METHODOLOGY

### *STUDY AREA*

Rhode Island is the smallest state in the United States with a total land area of 3,100 $km^2$. However, it is ranked among the top states for both human population density and road density (United States Census Bureau, 2010). Rhode Island harbors 18 species of native amphibians (10 salamanders, 8 frogs) and 19 species of native reptiles (12 snakes, 7 non-marine turtles), many of which are suspected to be undergoing population declines because of anthropogenic stressors including habitat loss and road mortality (Rhode Island Wildlife Action Plan, 2015). My study was conducted in western Rhode Island throughout Kent and Washington counties (Figure 1).

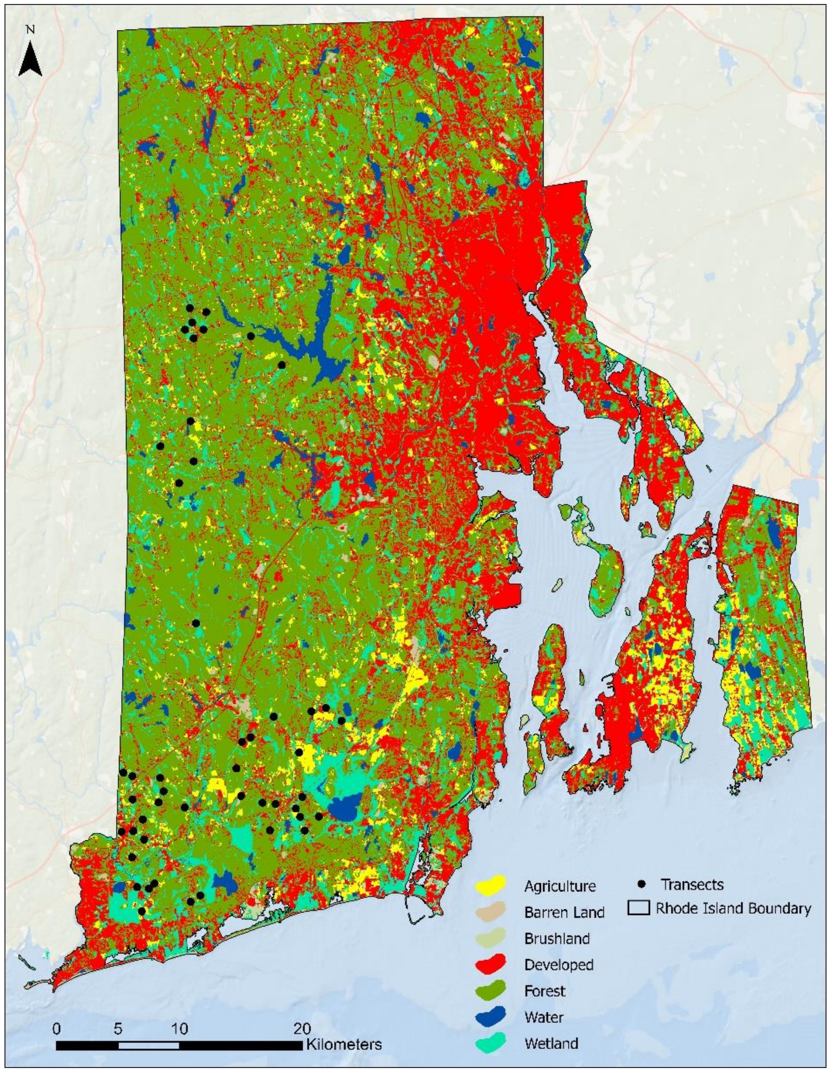


Figure 1. Land use and land cover and the location of study transects surveyed from 2019–2021 in Rhode Island.

*ROAD MORTALITY SURVEYS*

Road mortality surveys were conducted from late April to mid-July from 2019 to 2021, during active periods of amphibians and reptiles, on two-laned paved roads with varying traffic volumes ranging from 15–6,000 vehicles per day, some of which intersected habitats used by amphibians and reptiles. Surveys were conducted at 48 randomly selected 200-m transects that were separated by at least 500-m to reduce spatial autocorrelation. Surveys were conducted between the hours of 20:00–01:30. Half of the transects were located within 100-m of a wetland and half of them were located farther than 100-m from a wetland. On each survey night, a group of 6 transects were surveyed using walking survey methods. For each year, transects were surveyed five times during the survey period. Weather conditions varied across surveys, which included evenings with and without precipitation. Surveys began approximately one hour after sunset on nights with air temperatures ≥5.5℃.

A survey consisted of pairs of two or more surveyors walking the length of each transect, one on each side of the road, scanning the surface and adjacent area for live or dead amphibians and reptiles. At the end of the transect, surveyors switched sides and walked backed to the beginning of the transect, continuing to scan for live or dead animals. I used a mobile application, Herp Observer Road Edition, developed by the Rhode Island Department of Environmental Management to document amphibians and reptiles found on or adjacent to the road along each transect. This application was developed within the data collection platform Survey123 (ESRI, Redlands, CA, USA) and was used to record the species, date, time, geographic coordinates, condition (alive or dead-on road), and any notes on the observation, along with a photo of each carcass. Carcasses that were highly decomposed were recorded at the genus level. Once recorded, carcasses were removed to reduce double counting on the return walk during each survey. Any live amphibians and reptiles encountered on the road during surveys were recorded and moved off the road in the direction they were headed. Environmental conditions during surveys were recorded using the North American Amphibian Monitoring Program protocol (Weir et al., 2005). Temperature was recorded during surveys and daily precipitation was obtained from the nearest Community Collaborative Rain, Hail, & Snow Network rain gauge (Colorado Climate Center, 2021).

*OCCUPANCY MODEL DEVELOPMENT*

In this study, I defined occupancy as the probability of a large roadkill event occurring along a transect. Detection probability was defined as the probability of detecting a large roadkill event during a survey given that one occurred along the transect. I defined a large roadkill event as observing five or more amphibians and/or reptiles dead at a transect during a survey. Five or more carcasses represented the top 10% of transects with the highest roadkill observations per survey. I considered testing a second scenario for roadkill events of ≥10 carcasses being observed along a transect during a survey, however, roadkill events of ≥10 individuals were only detected at 11.7% (n=7) of all transects. Given the infrequency of these events, reliable estimates using the occupancy modeling approach was not feasible. Repeated visits to a site are required when using occupancy modeling to determine detection history, which is a pattern of detection or non-detection of a species at a site, or in this case detection of a large roadkill event. Due to the low number of roadkill observations by taxonomic group, all amphibian and reptile roadkill were aggregated regardless of species at each transect. This decision was supported by findings which determined that road mortality hotspots for amphibians and reptiles overlapped (Langen et al., 2009). Live amphibians and reptiles on the road were documented but were not included in analyses. I used the ‘unmarked’ package in R (Version 4.1.1) to estimate occupancy and detection probability for large roadkill events. The ‘unmarked’ package uses maximum likelihood estimates to predict occupancy and detection probability from observed data for detection or non-detection of a species.

Geospatial data for Land Use and Land Cover (RIGIS, 2011) and Rhode Island Department of Transportation Roads (RIGIS, 2016) were acquired from the Rhode Island Geographic Information System. The Land Use and Land Cover data were classified using the Anderson Level III classification system (Anderson, 1976). In addition, I acquired data from the National Wetlands Inventory (USFWS, 2021). I used ArcMap (ESRI, ArcMap version 10.1, Redlands, CA) to merge land use and land cover data with the National Wetlands Inventory data. Percentage land cover surrounding a road was calculated by buffering each transect by 100-m and 500-m and dividing the area of each land use class within the buffer by the total buffer area. Scales of 100-m and 500-m were chosen as they have been identified as being the scales associated with amphibian and reptile road morality hotspots (Langen et al., 2009). Current comprehensive traffic volume data were not available within the state, so I used Functional Class Description Codes from the Federal Highway Administration (FHWA, 2013) as a proxy for traffic volume levels, which assigns classes to roads based on traffic volume ranges (Table 1).

Table 1: Federal Highway Administration functional class description codes used to estimate traffic volume for roads in Rhode Island.

| Code | Functional Class Description | Traffic Volume (Vehicles/ Day) |
|---|---|---|
| 1 | Interstate | 12,000–34,000 |
| 2 | Other Freeways & Expressways | 4,000–18,500 |
| 3 | Other Principal Arterial | 2,000–8,500 |
| 4 | Minor Arterial | 1,500–6,000 |
| 5 | Major Collector | 300–2,600 |

| | | |
|---|---|---|
| 6 | Minor Collector | 150–1,110 |
| 7 | Local | 15–400 |

Using roadkill observation data from surveys, I generated a detection history matrix (Table 2) for each transect surveyed to determine if a large roadkill event was detected (1) or not detected (0) during each survey.

Table 2: Example of the detection history matrix used to code for large roadkill events along surveyed road transects in Rhode Island.

| **Transect** | **Survey 1** | **Survey 2** | **Survey 3** | **Survey 4** | **Survey 5** |
|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 1 |
| 2 | 1 | 1 | 0 | 0 | 0 |
| 3 | 0 | 0 | 0 | 0 | 0 |
| 4 | 0 | 1 | 1 | 0 | 0 |

Using the survey detection history, I developed a model that assumed an unvarying influence of site and survey characteristics to estimate constant occupancy and detection probability across all transects. Then, I developed a model set for each scale (100-m and 500-m) consisting of a combination of different covariates (Table 3) to estimate occupancy and detection probability for all transects. Covariates for occupancy included percent wetland area and water within a 100-m and 500-m buffer around a transect. The Anderson Level III system classifies wetlands as forested and non-forested wetlands and water as streams, lakes, reservoirs, and bays (Anderson, 1976). Roads were classified using the Functional Class Description Codes defined in

Table 1. For detection probability, three observational covariates were evaluated, including temperature during each survey, 24-hour precipitation recorded from the nearest rain gauge, and day of year starting from 1 January of the survey year.

Table 3: Covariates evaluated for occupancy and detection probability using the modeling framework.

| Covariate | Definition | Data Type |
| --- | --- | --- |
| Perc_Water_100m | % Water (streams and lakes) within 100-m of a road | Continuous |
| Perc_Wetland_100m | % Wetland (forested and non-forested wetlands) within 100-m of a road | Continuous |
| Perc_Water_500m | % Water within 500-m of a road | Continuous |
| Perc_Wetland_500m | % Wetland within 500-m of a road | Continuous |
| F_Class_Code | Road classification | Categorical |
| AvgTemp | Temperature recorded during surveys | Continuous |
| DayRain | Precipitation in previous 24 hours recorded at nearest weather gauge | Continuous |
| TimeOfYear | Day of the year from 1 January | Continuous |

I limited the number of covariates included in the model to those that I expected to influence road mortality based on published literature with the goals of maximizing reliability of model estimates and reducing complexity of the models. Each model set consisted of 15 sub-models. Models for each model set were compared using Akaike's Information Criterion, a metric from information theory in which a set of candidate models is developed *a priori* based on prior knowledge and models are evaluated based on model fit and complexity (Anderson & Burnham, 2004). I used model averaging to make predictions of occupancy and detection probability for all transects. Model averaging considers the variation in estimates among models allowing for more reliable predictions (Anderson & Burnham, 2004). Finally, using the simplest model that assumed constant occupancy and detection probability across all transects, I determined the survey effort needed to estimate the probability of detecting a large roadkill event at least once at occupied transects as:

$$p^{*} = 1\text{-}(1\text{-}p)^{s},$$

where (p*) is the probability of detecting at least one large roadkill event at a transect assuming it occurs there, (p) is the estimated per survey detection probability, and (s) is the number of surveys. From this, I was able to estimate the number of surveys that needed to be conducted to have a probability of 0.85 and 0.95 of detecting a large roadkill event to guide survey efforts (i.e., number of surveys) needed to ensure a large roadkill event was present at a transect.

## RESULTS

### *ROAD MORTALITY SURVEYS*

I conducted 240 surveys between April 2019 to July 2021, in which a total of 60 km of roadway were surveyed. Mean temperature during surveys was 15.1 °C (SD = 5.63, range = 5.6–22.6 °C) and mean daily precipitation in the preceding 24 h was 0.31 cm (SD = 0.60, range = 0–3.3 cm). I recorded 639 live and dead amphibians and reptiles, of which 95% were found dead on the road. This equated to a roadkill rate of 10.1 carcasses per km/day. Of the 606 roadkill observations, 19% were too damaged to be identified at the genus or species level. The largest roadkill event we observed while surveying was 33 carcasses. Nineteen native species were observed dead on roads, of which 82% were frogs, 7 % were salamanders, 5% were snakes, and 6% were turtles (Table 3).

Table 4. Live and dead amphibians and reptiles observed on roads in Rhode Island 2019–2021.

| Group | Dead on road | Live on road | Total | % Dead |
|---|---|---|---|---|
| **Frog** | 496 | 30 | 526 | 94 |
| **Salamander** | 44 | 3 | 47 | 94 |
| **Snake** | 27 | 0 | 27 | 100 |
| **Turtle** | 39 | 0 | 39 | 100 |

The most frequently observed amphibians (Table 5) were the American toad (*Anaxyrus americanus*) and wood frog (*Lithobates sylvaticus*). The most frequently observed reptiles (Table 5) were the ring-necked snake (*Diadophis punctatus*) and Eastern painted turtle (*Chrysemys picta*).

Table 5. Distribution of mortality by species for roadkill surveys conducted in Rhode Island, 2019–2021.

| Taxonomic Group/Species | Scientific Name | % of Observations |
|---|---|---|
| **Frog** | | |
| American toad | *Axaxyrus americanus* | 14% |
| American bullfrog | *Lithobates catesbeianus* | 1% |
| Fowlers toad | *Anaxyrus fowleri* | < 1% |
| Gray treefrog | *Hyla versicolor* | 10% |
| Green frog | *Lithobates clamitans* | 15% |
| Pickerel frog | *Lithobates palustris* | 2% |
| Spring peeper | *Pseudacris crucifer* | 11% |

| | | |
|---|---|---|
| Wood frog | *Lithobates sylvaticus* | 14% |
| Unidentified *Lithobates* | - | 7% |
| Unidentified *Anaxyrus* | - | 1% |
| **Salamander** | | |
| Four-toed salamander | *Hemidactylium scutatum* | 1% |
| Eastern Red-backed salamander | *Plethodon cinereus* | 6% |
| Spotted salamander | *Ambystoma maculatum* | < 1% |
| Northern Two-lined salamander | *Eurycea bislineata* | < 1 % |
| **Snake** | | |
| Common Gartersnake | *Thamnophis sirtalis* | 1% |
| Eastern Milksnake | *Lampropeltis triangulum* | 1% |
| Northern watersnake | *Nerodia sipedon* | < 1% |
| Southern Ring-necked snake | *Diadophis punctatus* | 2% |
| **Turtle** | | |
| Eastern painted turtle | *Chrysemys picta* | 4% |
| Snapping turtle | *Chelydra serpentina* | 1% |
| Spotted turtle | *Clemmys guttata* | 1% |
| **Unidentified carcass** | - | 10% |


Figure 2. Examples of amphibians and reptiles observed dead on roads in Rhode Island in 2019–2020. A) Green frog (*Lithobates clamitans).* B) Eastern red-backed Salamander (*Plethodon cinereus*); C) Ring-necked snake (*Diadophis punctatus*); Painted turtle (*Chrysemys picta*).

*OCCUPANCY MODELING*

I found that the percentage of wetland cover within 100-m of a road and traffic volume (as estimated by road classification) were the most supported covariates

(Table 6), positively influencing estimated site occupancy at the 100-m scale (Figure 3). Specifically, estimated occupancy was highest at transects on roads with low traffic volume (15–400 vehicles/day). I found that precipitation in the preceding 24-h and road classification most strongly influenced detection probability at the 100-m scale (Table 6). Detection probability increased ($p < 0.001$) with daily precipitation and decreased ($p < 0.001$) as traffic volume decreased (Figure 4).

Table 6. Occupancy and detection estimates from the top model in the 100-m model set.

| **Parameters** | | | | |
|---|---|---|---|---|
| **Occupancy** | **Estimate** | **SE** | **Z** | **P(>\|z\|)** |
| Intercept | 3.01 | 2.17 | 1.39 | 0.165 |
| Percent wetland within 100-m | 1.24 | 1.18 | 1.05 | 0.292 |
| Road classification | 1.38 | 1.15 | 1.20 | 0.232 |
| | | | | |
| **Detection** | | | | |
| Intercept | -2.195 | 0.257 | -8.53 | <0.001 |
| Rain | 0.777 | 0.161 | 4.92 | <0.001 |
| Road classification | -0.790 | 0.228 | -3.47 | <0.001 |

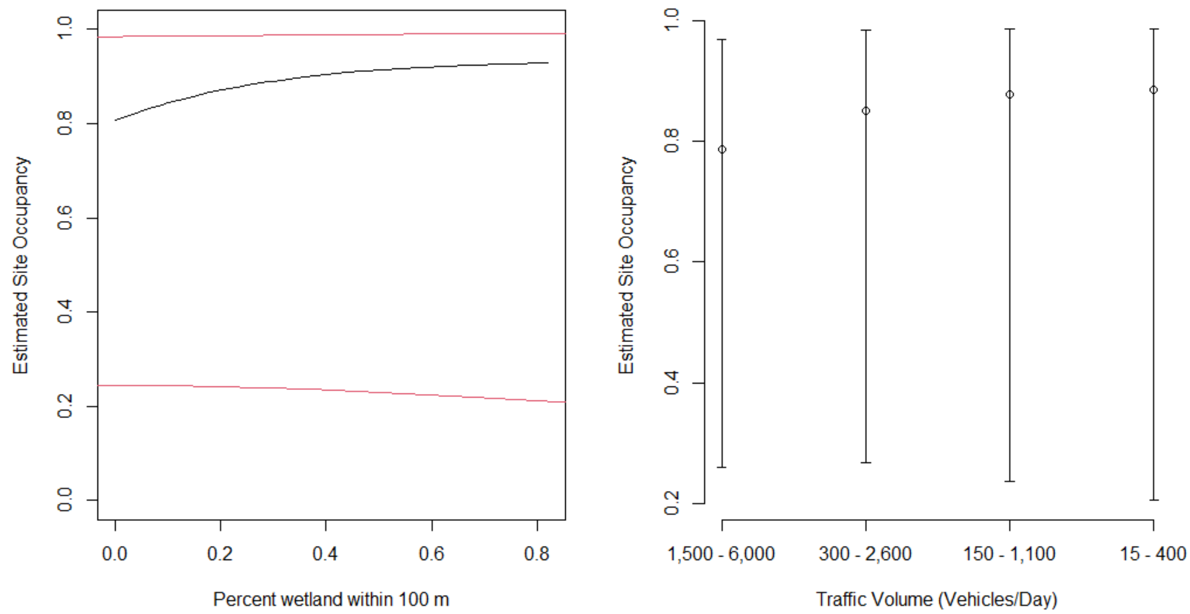


Figure 3: Marginal effects of covariates from the most supported model on estimated site occupancy in the 100-m scale model subset. Black lines represent model estimates for occupancy and red lines represent upper and lower confidence intervals. Plots hold other covariates at their mean value.

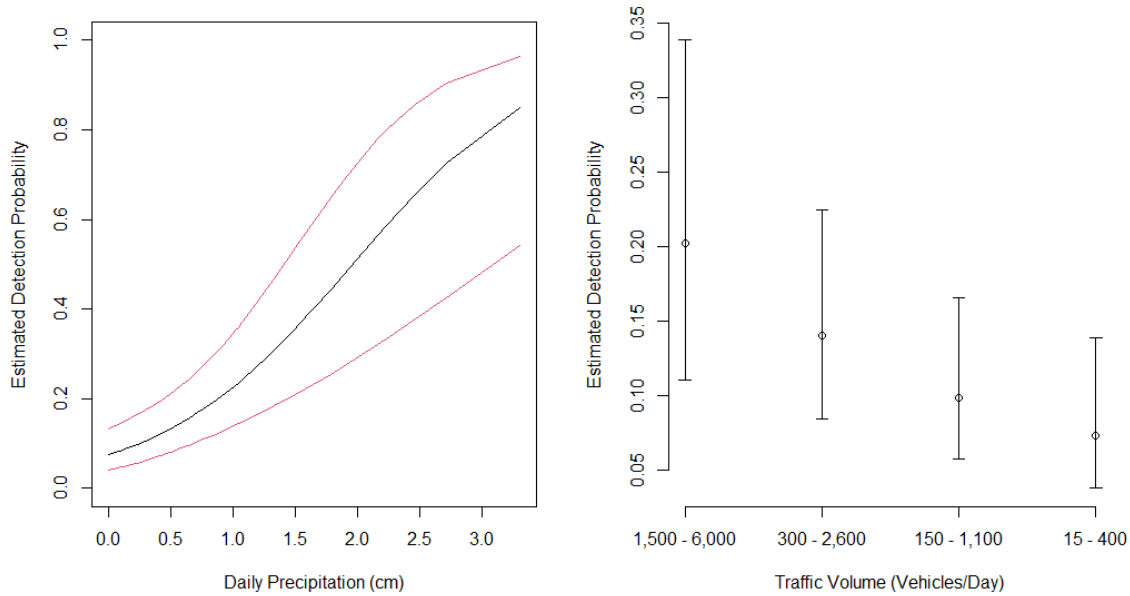


Figure 4: Marginal effects of covariates from the most supported model on estimated detection probability in the 100-m scale model subset. Black lines represent model estimates for occupancy and red lines represent upper and lower confidence intervals. Plots hold other covariates constant at their mean value.

In comparison, the 500-m model scale set results indicated that the percentage of area covered by water within a 500-m buffer of each transect and road classification were the most supported covariates (Table 7), with both having a positive effect on estimate site occupancy (Figure 5). Similar to results at the 100-m scale, site occupancy was highest on roads with low traffic volume (15–400 vehicles/day). For detection probability, daily precipitation and road classification were the most supported covariates (Table 7). Detection probability increased ($p < 0.001$) with daily precipitation and decreased ($p < 0.001$) as traffic volume decreased (Figure 6).

Table 7. Occupancy and detection estimates for the top model in the 500 m scale model set.

| **Parameters** | **Model 29** | | | |
|---|---|---|---|---|
| **Occupancy** | **Estimate** | **SE** | **Z** | **P(>\|z\|)** |
| Intercept | 2.590 | 1.613 | 1.61 | 0.108 |
| Percent water within 500-m | 2.311 | 1.692 | 1.37 | 0.172 |
| Road classification | 0.919 | 0.845 | 1.09 | 0.277 |
| | | | | |
| **Detection** | **Estimate** | **SE** | **Z** | **P(>\|z\|)** |
| Intercept | -2.108 | 0.274 | -7.70 | <0.001 |
| Rain | 0.768 | 0.165 | 4.65 | <0.001 |
| Road classification | -0.784 | 0.231 | -3.40 | <0.001 |

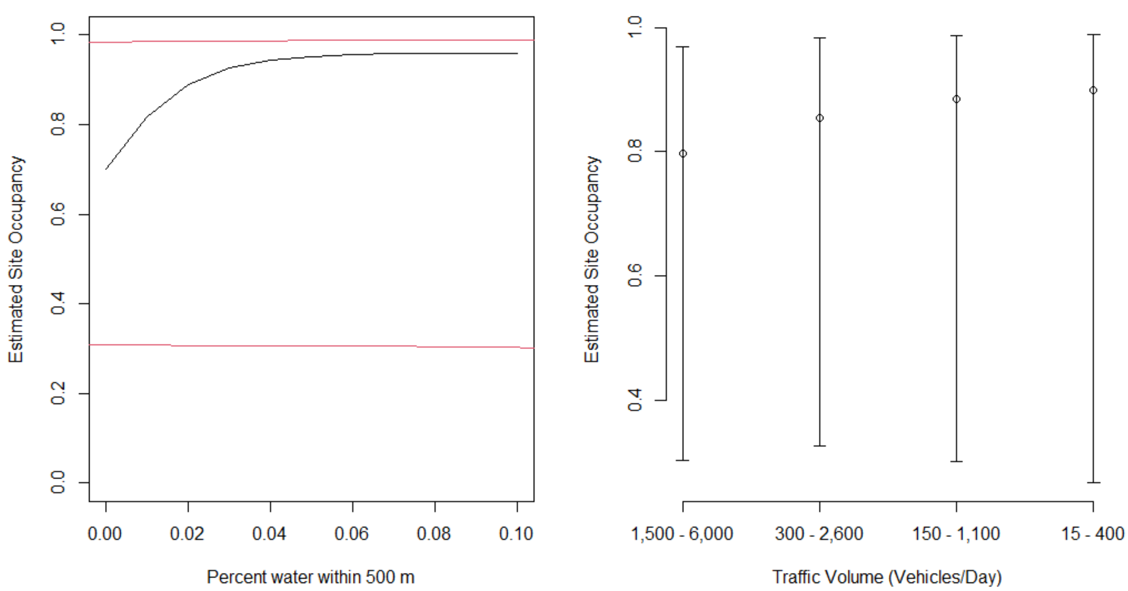


Figure 5. Marginal effect of covariates on estimated site occupancy in the 500-m scale model subset. Black lines represent model estimates for occupancy and red lines represent upper and lower confidence intervals. Plots hold other covariates constant at their mean value.

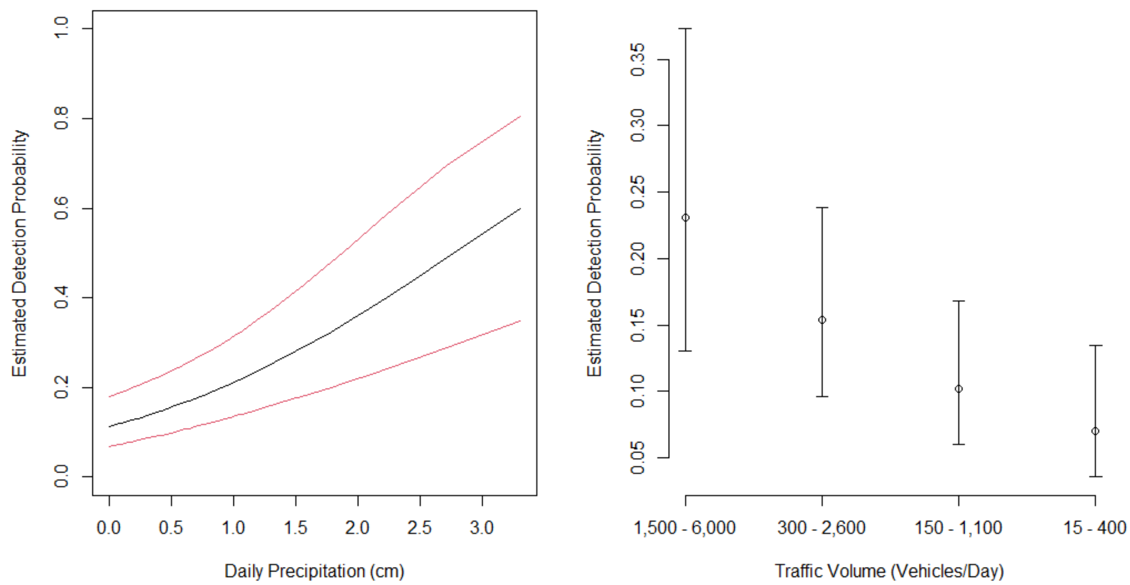


Figure 6: Marginal effect of covariates on estimated detection probability in the 500-m scale model subset. Black lines represent model estimates for detection probability and red lines represent upper and lower confidence intervals. Plots hold other covariates constant at their mean value.

At 40% of transects, I observed at least one large roadkill event, which represented the naïve occupancy or proportion of transects with at least one large roadkill event. Assuming constant occupancy and detection, I estimated a site occupancy of 0.64, meaning that a large roadkill event occurred at 64% of surveyed transects. I estimated a detection probability of 0.175, or a 17.5% chance of detecting a large roadkill at transect given it occurs there. Using these estimates, the survey effort required to determine if a large roadkill event occurs with 85% certainty is 10 surveys and 95% certainty is 16 surveys (Figure 7).

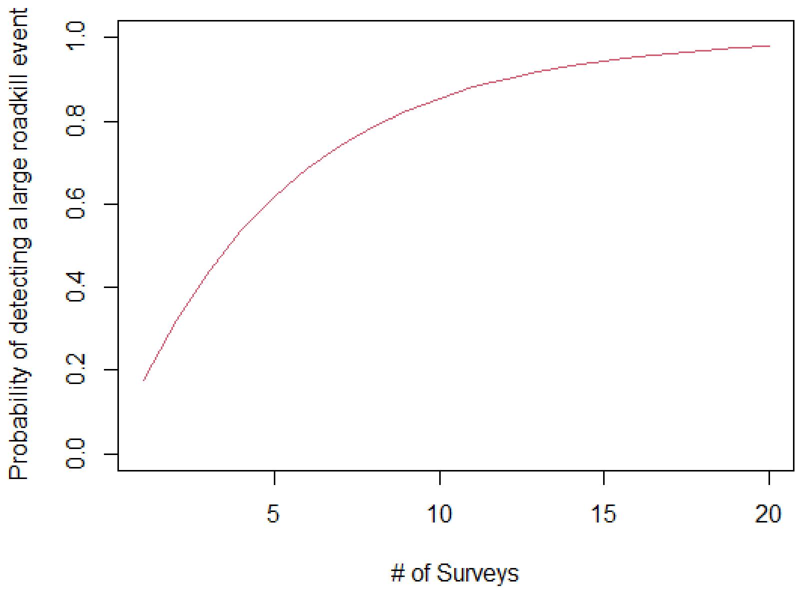


Figure 7. Number of surveys versus the probability of detecting at least one large roadkill event (≥5 amphibian or reptile carcasses) at occupied transects.

## DISCUSSION

### *ROAD MORTALITY SURVEYS*

This study was the first to assess amphibian and reptile road mortality in Rhode Island, a state undergoing rapid change that includes increasing human populations and associated road development. Overall, the roadkill rates I observed were similar to those of other studies addressing amphibian and reptile road mortality, which reported 2–8 carcasses/km/day (Glista et al., 2008; Langen et al., 2007). Similar to the results of others (e.g., Fahrig et al., 1995; Glista et al., 2008), amphibians made up the majority (89%) of all roadkill observations. This is likely due to their high abundance near roadways, in addition to having several life history characteristics that predispose them to road mortality, such as making frequent road crossings and not actively avoiding vehicles once in the road (Bouchard et al., 2009;

Mazerolle, 2004). Higher numbers of frog carcasses compared to salamander carcasses may reflect differences in abundances between the two groups or challenges with detection based on the small sizes of some salamanders. In addition, frogs tend to be more active and make more frequent road crossings. Reptiles represented a smaller proportion of all roadkill observations. Turtles and many snakes are diurnal and it is likely that their carcasses did not persist on the road long enough due to predation to be observed during my night surveys. Reptiles are also more active during the day and it is possible that were more incidences of road mortality, however, due to the timing of surveys, these carcasses were not observed. However, all reptiles observed in this study were found dead on the road, suggesting that there is a higher risk of road mortality for reptiles. This supports findings from other studies that observed a high risk of road mortality for reptiles crossing roads, especially for turtles (Aresco, 2005; Gibbs & Shriver, 2002). This is of particular concern as the low fecundity and delayed sexual maturity of many turtle species can make populations highly vulnerable to the impacts of road mortality, leading to local population declines and extirpation (Howell & Seigel, 2019). For example, at two transects that both intersected large wetland complexes, I observed multiple spotted turtle carcasses. The spotted turtle is a Species of Greatest Conservation Need in Rhode Island due to population declines (RIDEM), 2015) and is a candidate for listing under the federal Endangered Species Act.

*OCCUPANCY MODELING*

The scale at which habitat was assessed around road transects influenced estimates of occupancy and detection probabilities for large roadkill events. In our study area, larger amounts of wetland area surrounding a road increased the probability of large roadkill events at a smaller scale (100-m). At a larger scale (500-

m), higher areal coverage by water bodies, such as streams and lakes, increased the probability of large roadkill events. This supports findings from other studies (Glista et al., 2008; Langen et al., 2009) that documented increases in road mortality at road sections with wetlands and water bodies adjacent to or bisected by a road.

At both small (100-m) and large (500-m) scales, I found that roads with lower traffic volume (15–400 vehicles/day) were associated with the highest occurrence of large roadkill events. This is likely due to roads with less traffic occurring in less developed areas that contained more habitat for amphibians and reptiles, thereby supporting larger populations that make more frequent road crossings. These results are similar to findings from other studies that observed higher levels of amphibian mortality on roads with lower traffic volume (Fahrig et al., 1995; Sutherland et al., 2010). This may be due to several reasons. First, roads with higher traffic volume may result in carcasses being more quickly destroyed before they are observed during a survey. Second, many roads with high traffic volume in Rhode Island occur in areas with high urban development that lack habitat for amphibians and reptiles. In addition, lower abundances of amphibians have been observed near high traffic volume roads, which could contribute to lower levels of mortality (Gravel et al., 2012). Amphibians and reptiles may also avoid road crossing on those with high traffic due to the increased disturbance by vehicles.

For detection probability, my results indicated that at transects where large roadkill events occurred, the probability of detecting the event during a survey increased on roads with higher traffic volume. While the probability of detecting a large roadkill event increased with traffic volume, large roadkill events were less likely to occur on high traffic volume roads. In other words, at transects on roads with high traffic volume, there is a low probability that a large roadkill event occurs. However, should a large roadkill event occur, there is a high probability that it would be detected during a survey. This is likely due to the increased flow of traffic increasing the risk of road mortality. Regardless of scale, large roadkill events were more likely to be detected during surveys with higher precipitation in the preceding 24 hours. During periods of increased precipitation, amphibians and reptiles more frequently cross roads to breed and forage, thereby increasing the risk of mortality (Gravel et al., 2012). As indicated by our results, occupancy and detection probability varied at transects depending on the surrounding habitat and timing of surveys.

Although several studies have used occupancy modeling to examine roadkill risk (Santos et al., 2018), this study is among the first to use occupancy modeling to identify locations where large amphibian and reptile roadkill events are most likely to occur. Importantly, our study has addressed the influence of imperfect detection during surveys on spatial patterns of road mortality, a challenge noted in several studies attempting to identify road mortality hotspots using roadkill counts (Degregrio et al., 2011; Ogletree & Mead, 2020; Santos et al., 2011; Teixeira et al., 2013). The results of this study are specific to our study area. However, the modeling framework developed could be applied to other regions by those interested in better targeting mitigation measures for herpetofaunal road mortality. Using occupancy modeling, I was able to address imperfect detection during surveys to generate more reliable estimates of the occurrence of large roadkill events. As indicated by the results of our occupancy analyses, large roadkill events occurred at a greater number of transects

(64%) than were observed during road surveys (40%). This is likely due to the low probability (17.5%) of detecting a large roadkill event during a single survey, suggesting that imperfect detection is influencing our ability to detect large roadkill events. Such events are likely being missed, and several studies have indicated that this may be a factor limiting identification of road mortality hotspots (Langen et al., 2007; Teixeira et al., 2013). Factors such as timing of surveys (e.g., surveying on a dry evening vs. an evening when it rains) or missing carcasses due to their size or being destroyed by cars or scavenged, potentially contributed to imperfect detection of roadkill during our surveys (Ogletree & Mead, 2020; Santos et al., 2011; Slater, 2002). By addressing imperfect detection during surveys, I was able to identify locations most appropriate for mitigation measures that reduce road mortality.

## CONSERVATION IMPLICATIONS

Mitigations measures, including infrastructure that keeps amphibians and reptiles off roads, can be costly and are most effective when implemented at locations with the greatest risk of road mortality rather than highest roadkill totals (Santos et al., 2018; Santos et al., 2011). When implemented appropriately, mitigation measures can be highly effective in reducing road mortality (Aresco, 2005; Beebee, 2013; Gonçalves et al., 2018). Using the results of our models, locations can be targeted for implementing mitigation measures, such as road fencing, culverts, and road tunnels that deter amphibians and reptiles from attempting road crossings (Glista et al., 2009). In addition, installing road signage and lighting during key times of year will remind drivers to slow down and use caution and better see animals on the road (Glista et al., 2009). Where possible, gates can be installed and used to close roads during brief seasonal breeding events, with signage for drivers about a detour path.

To implement mitigation measures most effectively, conservation managers should consider the following approach when addressing herpetofaunal road mortality. I recommend conducting surveys on sections of road with low traffic near habitat for amphibians and reptiles, where large roadkill events are most likely to happen. By targeting surveys, conservation managers can better identify locations where road mortality is most likely to occur at high rates. However, the influence of imperfect detection on spatial patterns of road mortality (e.g., non-detection leading to false absence) can misguide the implementation of mitigations measures, reducing their effectiveness in preventing road mortality (Santos et al., 2015). As I have demonstrated, there is a low probability of detecting a large roadkill event due to imperfect detection during surveys. Although the probability of detecting a large roadkill event increases with survey effort, conducting more surveys may have marginal gains in identifying high-occurrence roadkill locations and are time- and resource-intensive (Shannon et al., 2014). In addition to survey effort, the timing of surveys is also important as surveying under ideal weather conditions better captures spatial patterns of road mortality (Santos et al., 2011). As our results indicate, precipitation has a strong influence on the probability of detecting a large roadkill event. Therefore, I recommend conducting surveys either during or immediately after a rain event, as large roadkill events are more likely to be detected. Using our approach, fewer surveys can be conducted and occupancy modeling can be used to address imperfect detection during surveys to identify and prioritize locations where mitigation measures would be most effective in reducing road mortality. Overall, the approach I have developed can be used for more cost-effective survey design by guiding survey effort (i.e., the number of surveys conducted) and when surveys should be conducted to better capture spatial patterns of large roadkill events. Importantly, our approach can be used to correct for imperfect detection during surveys to prioritize locations based on those with the highest probability of a large roadkill event occurrence, rather than targeting locations with the highest roadkill totals (Santos et al., 2018). Using this information, mitigation measures can be implemented that are more effective in preventing road mortality, thereby reducing the impacts of roads and traffic on amphibians and reptiles.

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